



Shri Guru Gobind Singhji Institute of Engineering and Technology, Nanded
ISE-II (Semester-II) AY:2025-26

Class: Second Year B. Tech.

Branch: EC/EL/TN/ME/PR

Course: Environmental Studies

Course Code: VEC-202

Date: 25-March-2026

Time: 45 Minutes (14:30 to 15:15)

Max. Marks: 20

Q. No.	Questions	Marks	CO	BT
Q. 1)	List out the different types of Ecosystem. Write a short note on desert ecosystem.	05	4	2
Q. 2)	Define biodiversity. Explain genetic, species and ecosystem diversities.	05	4	2
Q. 3)	What are the major threats to biodiversity? Explain briefly.	05	4	2
Q. 4)	Explain the following values of biodiversity: consumptive use value, productive use value, social value, ethical value and option value.	05	4	2

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Shri Guru Gobind Singhji Institute of Engineering and Technology, Nanded

ISE-II (Semester-II) 2025-26

Second Year B. Tech.

Civil Engineering

Course: Concrete Technology

Course Code: PCC-CE207

Date: 27-Mar-2026

Time: 45 Minutes (16:30 to 17:15)

Max. Marks: 20

Q. No.	Questions	Marks	CO	BT
Q. 1)	Define admixture. What are its needs? Enlist the types of admixtures and explain any two in detail.	10	CO2	BT2
Q. 2)	Enlist the various types of special concrete and explain any three in detail.	10	CO5	BT2

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Shri Guru Gobind Singhji Institute of Engineering and Technology, Nanded

ISE-II (Semester-II) 2025-26

Second Year B. Tech.

Civil Engineering

Course: Building Planning and Construction

Course Code: PCC-CE206

Date: 27-Mar-2026

Time: 45 Minutes (14:30 to 15:15)

Max. Marks: 20

Q. No.	Questions	Marks	CO	BT
Q. 1)	Give the examples of: a) Institutional Buildings, b) Assembly Buildings, c) Mercantile Buildings	03	1	2
Q. 2)	Write a short note on below given basic requirements of a building: a) Strength and Stability, b) Dimensional Stability, c) Comforts and Convenience.	06	2	2
Q. 3)	Mr. John plans to construct a house in an area with high wind velocity and seismic activity. Which structural, material, and component options would you recommend to him for a stable and durable house?	07	2	5
Q. 4)	How are the lateral earth pressures acting on the retaining wall predicted by soil exploration method?	04	1	4

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Civil
Shri Guru Gobind Singhji Institute of Engineering and Technology, Nanded
Department of Civil Engineering
ISE-II Examination (Semester - II)
Academic Year: 2025-26

Class: SY-B.Tech.
(CE/CSE/IT/CH/TEXT)

Subject: Project Management

Subject Code: HSSM-201

Date: 25-03-2026

Time: 14:30 pm to 15:15 pm

Max. Marks: 20

Note:

- i. *Figures to right indicate full marks.*
- ii. *Use nonprogrammer calculator is permitted.*

ii. Use nonprogrammer calculator is permitted.

Q.No.	Question	Marks	CO	BT																														
Q1	Define Project Appraisal and explain its objectives and importance in project planning.	05	CO1	BT1																														
Q2	Explain the Payback Period method for project evaluation with advantages and limitations.	05	CO2	BT2																														
Q3	A company is considering the purchase of a new machine. Two alternative machines (X and Y) have been suggested, each having an initial cost of 20,00,000 and requiring 1,00,000 as additional working capital at the end of 1st year. Earnings after taxation are expected to be as follows. <table><tr><th>Year</th><th>Machine-X</th><th>Machine-Y</th></tr><tr><td>1</td><td>2,00,000</td><td>6,00,000</td></tr><tr><td>2</td><td>6,00,000</td><td>8,00,000</td></tr><tr><td>3</td><td>8,00,000</td><td>10,00,000</td></tr><tr><td>4</td><td>12,00,000</td><td>6,00,000</td></tr><tr><td>5</td><td>8,00,000</td><td>4,00,000</td></tr></table> <table><tr><th>Year</th><th>1</th><th>2</th><th>3</th><th>4</th><th>5</th></tr><tr><td>PV</td><td>0.91</td><td>0.83</td><td>0.75</td><td>0.68</td><td>0.62</td></tr></table> The company has target for a return of capital of 10% and on this basis, you are required to prepare the profitability of the machines and state which alternative you consider financially preferable using NPV method.	Year	Machine-X	Machine-Y	1	2,00,000	6,00,000	2	6,00,000	8,00,000	3	8,00,000	10,00,000	4	12,00,000	6,00,000	5	8,00,000	4,00,000	Year	1	2	3	4	5	PV	0.91	0.83	0.75	0.68	0.62	05	CO3	BT5
Year	Machine-X	Machine-Y																																
1	2,00,000	6,00,000																																
2	6,00,000	8,00,000																																
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5	8,00,000	4,00,000																																
Year	1	2	3	4	5																													
PV	0.91	0.83	0.75	0.68	0.62																													
Q4	Explain the TOPSIS method used in Project selection	05	CO2	BT2																														

== ALL THE BEST ==

Bloom's Taxonomy (BT)

BT1 - Remember, BT2 - Understand, BT3 - Apply, BT4 - Analyse, BT5 - Evaluate, BT6 - Create

Course Outcomes:

CO	On successful completion of this course, students should be able to:
CO 1	Appraise the selection and initiation of individual projects and its portfolios in an enterprise.
CO 2	Analyze the project planning activities that will predict project costs, time schedule, and quality.
CO 3	Develop processes for successful resource allocation, communication, and risk management.
CO 4	Evaluate effective project execution and control techniques that results in successful project completion



S. G. G. S. Institute of Engineering and Technology
Vishnupuri, Nanded-431606, India
Department of Civil Engineering

IN-SEMESTER EXAMINATION 2 – 2025-26 (Semester-II)

Date: Monday, 23-March-2026

Course: Theory of Structures – I

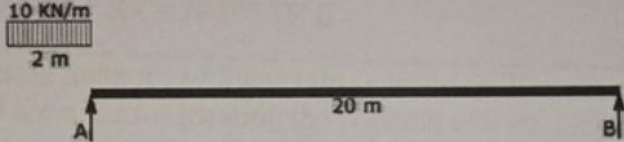
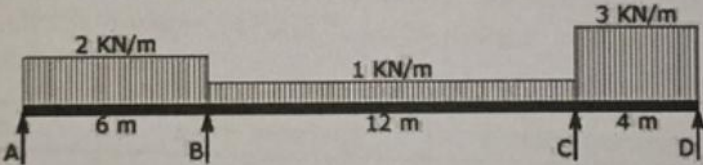
Time: 14:30 – 15:15

Class: S.Y. Civil Engineering

Course Code: PCC-CE205

Maximum Marks: 20

- Note**
1. Attempt all questions. Read question statement carefully.
 2. Use of non-programmable calculator is permitted
 3. Assume appropriate data, if necessary. Draw necessary diagrams wherever required.

Que.	Question statement	Mark	CO	BT
Q.1	A uniformly distributed load (UDL) rolling on the girder of span of 20 m from left to right. The UDL load is 2.0 m long and having load of 10 kN per meter. Determine, i. Maximum Bending moment on girder ii. Maximum shear force on girder iii. Determine the maximum positive and negative shear force at 5 m from the left support. 	10	CO3 CO4	BT4
Q.2	A cantilever beam ABCD covers three spans, AB = 6 m, BC = 12 m and CD = 4 m. It carries uniformly distributed loads of 2 kN/m, 1 kN/m and 3 kN/m on AB, BC and CD respectively. The girder is of same cross-section throughout. Determine the bending moment at the supports B and C and the pressure on each support. Plot the bending moment and shear force diagrams. 	10	CO2 CO3	BT4

Course Outcome (CO)	Bloom's Taxonomy Level (BT)
CO1: Apply Castigliano's theorems and the conjugate beam method to determine deflections in beams and trusses.	BT1 – Remember
CO2: Analyse fixed beams for internal forces and moments, including the effects of support settlement.	BT2 – Understand
CO3: Evaluate continuous beams and frames using Clapeyron's theorem of three moments.	BT3 – Apply
CO4: Construct and interpret ILDs for reactions, shear forces, and bending moments in beams and trusses.	BT4 – Analyse
CO5: Analyse long columns subjected to eccentric and lateral loads, including initial curvature.	BT5 – Evaluate
CO6: Determine cable forces and internal stresses in three-hinged suspension bridges under various loadings.	BT6 – Create

* * * All the Very Best * * *



Shri Guru Gobind Singhji Institute of Engg. & Technology, Vishnupuri, Nanded-431606.

Department of Civil Engineering
In-Semester Examination / Evaluation II
Academic Year: 2025-26 (Semester II)

Class: Second Year B.Tech.

Course: Surveying &
Geomatics

Course Code: PCC-CE204

Date: 24.03.2026

Time: 14:30 to 15:15

Max. Marks: 20

Note:

- Assume suitable data whenever necessary and state it clearly.
- Figures to right indicate full marks.

Q.N	Questions	Marks	CO	BT																		
Q.1	The following are the Fore Bearing of given traverse, find the Back Bearings: - a) FB of AB = $78^{\circ} 45'$ b) FB of BC = $192^{\circ} 15'$ c) FB of CD = $320^{\circ} 30'$ d) FB of DE = S $65^{\circ} 25'$ W e) FB of EA = N $30^{\circ} 15'$ E	05	CO2	BT2																		
Q.2	List the different methods of Plane table surveying. Describe any two methods that are used for locating the position of a station, with neat sketches.	05	CO3	BT3																		
Q.3	State the uses and advantages of a transit theodolite over a compass in civil engineering project.	04	CO2	BT2																		
Q.4	The following are the observed Fore Bearing (FB) and Back Bearing (BB) of a closed traverse PQRSTP, taken using a compass. Local attraction is suspected. <table><tr><th>Line</th><th>Fore Bearing</th><th>Back Bearing</th></tr><tr><td>PQ</td><td>$110^{\circ} 20'$</td><td>$289^{\circ} 40'$</td></tr><tr><td>QR</td><td>$35^{\circ} 10'$</td><td>$214^{\circ} 50'$</td></tr><tr><td>RS</td><td>$210^{\circ} 30'$</td><td>$30^{\circ} 30'$</td></tr><tr><td>ST</td><td>$300^{\circ} 00'$</td><td>$120^{\circ} 00'$</td></tr><tr><td>TP</td><td>$25^{\circ} 40'$</td><td>$204^{\circ} 50'$</td></tr></table>	Line	Fore Bearing	Back Bearing	PQ	$110^{\circ} 20'$	$289^{\circ} 40'$	QR	$35^{\circ} 10'$	$214^{\circ} 50'$	RS	$210^{\circ} 30'$	$30^{\circ} 30'$	ST	$300^{\circ} 00'$	$120^{\circ} 00'$	TP	$25^{\circ} 40'$	$204^{\circ} 50'$	06	CO3	BT3
Line	Fore Bearing	Back Bearing																				
PQ	$110^{\circ} 20'$	$289^{\circ} 40'$																				
QR	$35^{\circ} 10'$	$214^{\circ} 50'$																				
RS	$210^{\circ} 30'$	$30^{\circ} 30'$																				
ST	$300^{\circ} 00'$	$120^{\circ} 00'$																				
TP	$25^{\circ} 40'$	$204^{\circ} 50'$																				

=== ALL THE BEST ===

Bloom's Taxonomy (BT)

BT1 - Remember, BT2 - Understand, BT3 - Apply, BT4 - Analyse, BT5 - Evaluate, BT6 - Create

Course Outcomes:

	On successful completion of this course, students should be able to:
CO 1	Understand the principles of surveying and techniques for basic and modern surveying.
CO 2	Demonstrate the knowledge of remote sensing, geomatics, and photogrammetry techniques for civil engineering problems.
CO 3	Apply different instruments, tools, applications, and techniques to determine area, positions, distances, and angles between two points on the earth's surface.
CO 4	Explain the concepts of modern surveying techniques and instrumentation.
CO 5	Analyse field conditions and implement the techniques for setting out alignments, curves, and contours.



Shri Guru Gobind Singhji Institute of Engg. & Technology, Vishnupuri, Nanded-431606.
Department of Civil Engineering
Academic Year: 2025-26 (semester II)
In-Semester Examination II

Class: SY

Course Name: Hydraulic Engineering

Course Code: PCC-CE206

Date: 28.03.2026

Time: 14:30 to 15:15

Max. Marks: 20

Note:

- Assume suitable data whenever necessary and state it clearly.
- Mobile phones and programmable calculators are strictly prohibited.
- Exchange/Sharing of stationery, calculator etc. not allowed.
- Figures to right indicate full marks.
- The symbols carry usual meaning.

Q.N.			Marks	CO	BT
Q.1	a	Define : (i) Momentum thickness ; (ii) Displacement thickness ; (iii) Drag Force ; (iv) Stream-lined body ; (v) Bluff body	05	CO3	BT1
	b	Find the ratio of displacement thickness to momentum thickness and momentum thickness to energy thickness for the velocity distribution in the boundary layer given by $\frac{u}{U} = 2\left(\frac{y}{\delta}\right) - \left(\frac{y}{\delta}\right)^2$ Where, u = velocity in boundary layer at a distance y U = Free stream velocity δ = Boundary layer thickness	05	CO3	BT3
Q.2	a	The pressure difference Δp in a pipe of diameter D and length l due to viscous flow depends on the velocity V, viscosity μ and density ρ . Using Buckingham's π theorem, obtain an expression for Δp .	06	CO4	BT3
	b	Describe in details Reynold's model law.	04	CO4	BT2

=== ALL THE BEST ===

Bloom's Taxonomy (BT)

BT1 - Remember, BT2 - Understand, BT3 - Apply, BT4 - Analyze, BT5 - Evaluate, BT6 - Create

Course Outcomes:

	On successful completion of this course, students should be able to:
CO 1	Apply the knowledge of fluid mechanics to address the uniform flow problems in open channels
CO 2	Solve flow problems of turbulent flow, pipe network analysis and water hammer.
CO 3	Recognize concept of laminar and turbulent boundary layer and its analysis.
CO 4	Perform dimensional and model analysis for different fluid flow problems
CO 5	Understand the functions and working principles of hydraulic machineries



Shri Guru Gobind Singhji Institute of Engg. & Technology, Vishnupuri, Nanded-431606.
Department of CE/TEXT/CSE/EXTC/INFT/INS/MECH
Academic Year: 2025-26 (semester II)
In-Semester Examination I

Class: SY. Course Name: Environmental Impact Assessment Course Code: OE-IS-CE202
Date: 30 March 2026 Time: 14:30 to 15:15 Max. Marks: 20

Note:

- i. Assume suitable data whenever necessary and state it clearly.
- ii. Mobile phones and programmable calculators are strictly prohibited.
- iii. Exchange/Sharing of stationery, calculator etc. not allowed.
- iv. Figures to right indicate full marks.
- v. The symbols carry usual meaning.

Q.N.		Marks	CO	BT
1	Discuss the objectives of EIA in brief.	5	CO1	BT1
2	Explain the steps in EIA with flowchart.	10	CO1	BT1
3	Define environmental auditing and list out the objectives.	5	CO2	BT3

=== ALL THE BEST ===

Bloom's Taxonomy (BT)

BT1 - Remember, BT2 - Understand, BT3 - Apply, BT4 - Analyze, BT5 - Evaluate, BT6 - Create

Course Outcomes:

	On successful completion of this course, students should be able to:
CO 1	To develop an understanding of current EIA method and the techniques and tools used.
CO 2	To develop an understanding of current assessment method and legislation.
CO 3	To understand the climate change phenomenon and its related issues on water, irrigation and its social implications.
CO 4	To apply knowledge acquired to the process of environmental impact modelling and predication as a design tool with application to a number of case studies.